



Murata Ferrite Beads For Mobility

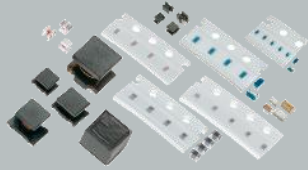
Feb. 2024

Product Line-Up

Capacitors



Inductors (Coils)



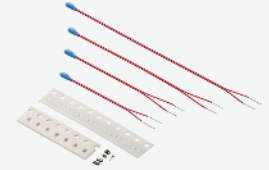
Noise/ EMI Suppression Products



Resistors



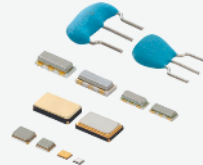
Thermistors



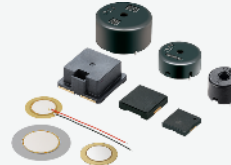
Sensors



Timing Devices



Sound Components



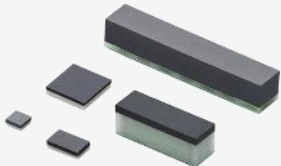
Power Devices



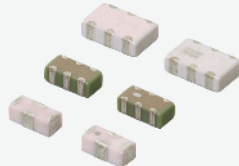
Micro Mechatronics



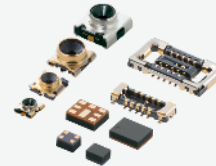
RFID Products



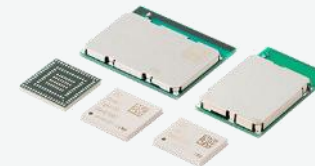
Matching Devices



RF Components



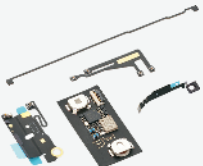
RF Modules



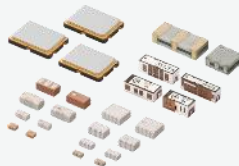
Batteries



High Frequency PCBs



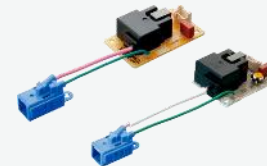
Filters



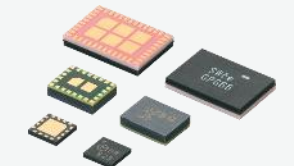
Antennas



Ionizers/ Ozonizers



Front End Modules



EMI Filters:

Ferrite Beads

BLM Series



Cost effective, Compact

Common Mode Choke Coils

DLW/ DLM Series



Excellent attenuation for signals and power

LC Filters

NFL/ NFE/ BNX Series



High power, High attenuating

Chip Inductors:

Power Inductors

DFE Series



Metal composite,
High performance

LQM Series



Multilayer type

LQH Series



Wire wound type

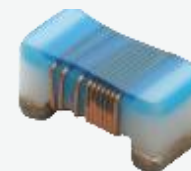
MDH Series



Wire
wound type

RF Inductors

LQW Series



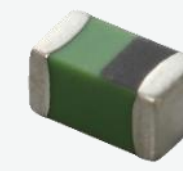
High Q Value

LQP Series



Compact

LQG Series



Cost effective

Automotive Grade Policy

Grade		Consumer 	Infotainment 	Powertrain/Safety  
Ferrite Bead		BLM15AG100S N 1	BLM15AG100S Z 1	BLM15AG100S H 1
CMCC		DLW21S N 121HQ2	DLW21S Z 121HQ2	-
Power inductor		DFE252012P-1R0M	DFE252012P D -1R0M	DFE2HCAH1R0MJ0
		LQH43P N 2R2M26	LQH43P Z 2R2M26	LQH43P H 2R2M26
		LQM21P N 2R2NGC	LQM21P Z 2R2NGC	LQM21P H 2R2NGC
RF inductor		LQP03TN1N0B O 2	LQP03TN1N0B Z 2	-
		LQG15H N 1N0B02	LQG15H Z 1N0B02	LQG15H H 1N0B02
		LQW18AN2N2D0 O	LQW18AN2N2D0 Z	-
AEC-Q200 specification		Not compliant	Compliant	Compliant
IATF16949 (former-ISO/TS16949)		ISO9001	Depend on products	IATF16949
Application		Non-Automotive	Passenger Compartment and information (Including IVI/Telematics)	Powertrain and Safety (Including ADAS)
Max. Operating Temp.		85/105/125degC	85/105/125degC	105/125/ 150degC or more
Specifications	High Temp. Exposure	1000 hr at Max. temp (*)	1000 hr at Max. temp	1000 hr at Max. temp
	Biased Humidity	1000h at 40degC / 95%RH Applied max rated current (*)	1000h at 85degC / 85%RH Applied max rated current	1000h at 85degC / 85%RH Applied max rated current
	Temp. Cycling	100 cycles Min. temp <-> Max. temp	1000 cycles Min. temp <-> Max. temp	1000 cycles Min. temp <-> Max. temp
Process Management		Standard	Strict	Target to Zero Defect

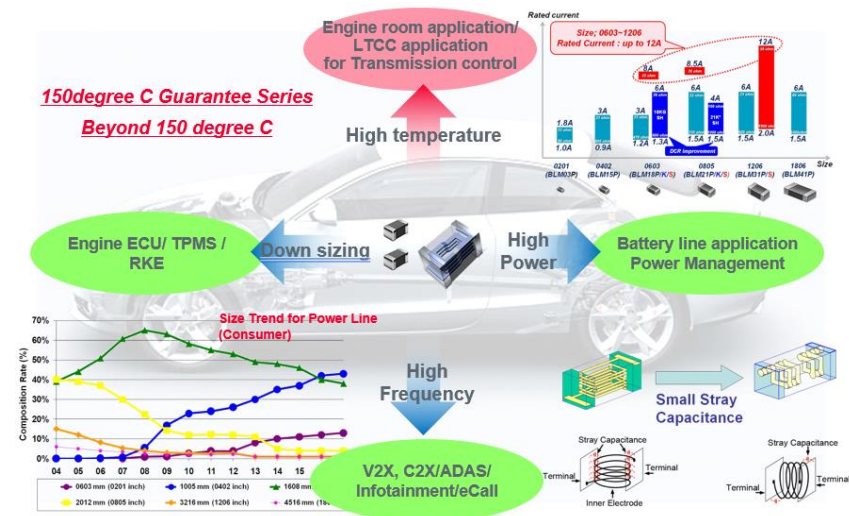
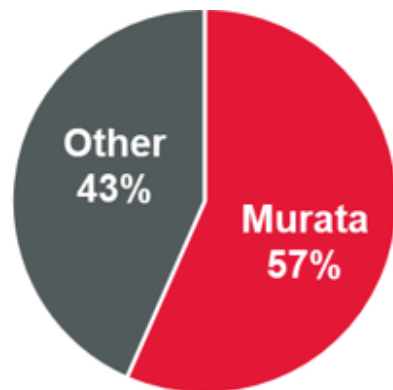
(*) : just example, actual test conditions (time, cycle, etc...) are different by product.

Chip Ferrite Beads

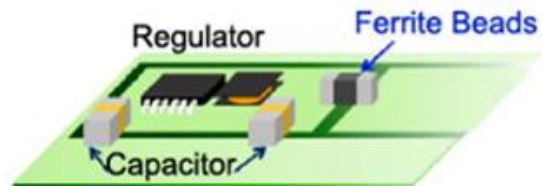


Overview of Ferrite beads

Series name : **BLM**



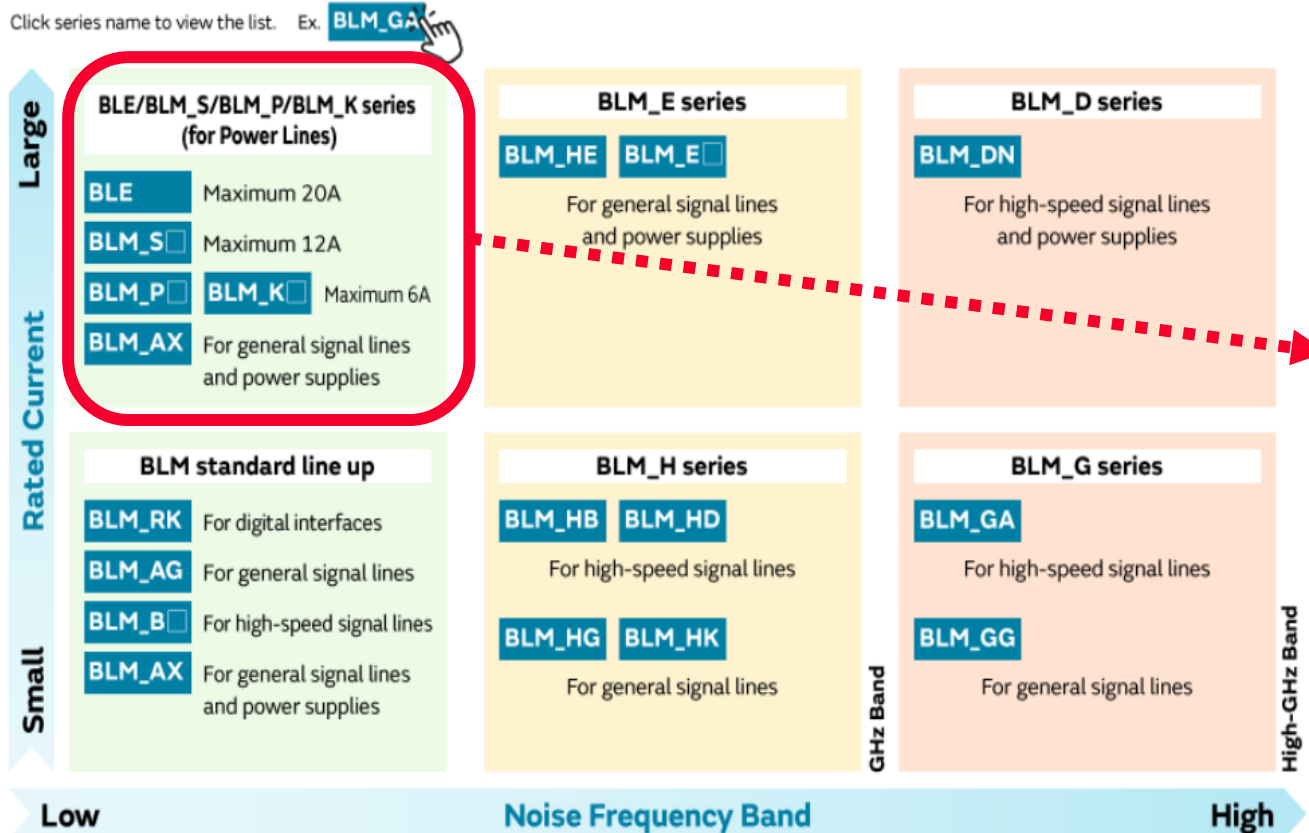
For power supplies



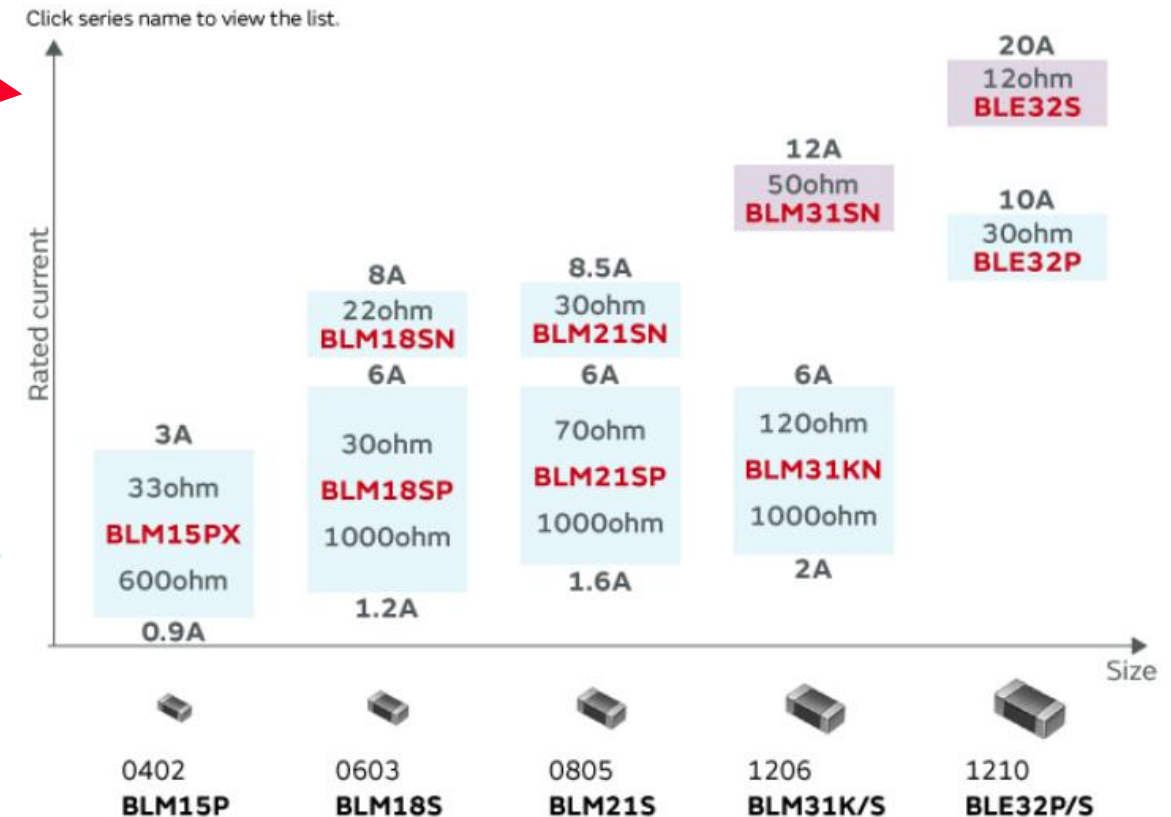
For signals



Focus Segment on Ferrite Beads

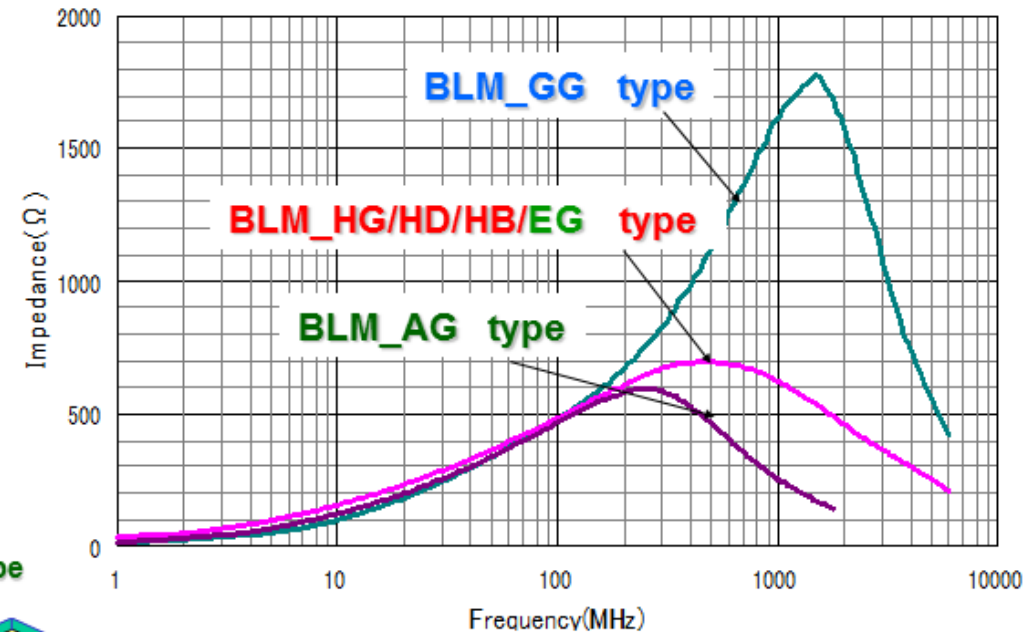


We have assembled a lineup of ferrite beads that support large currents as measures against power line noise. In particular, we recommend our **BLE32S** series, which supports currents **up to 20A**, the **BLM31SN** series, which supports currents **up to 12A**, and other similar series as products for power circuits that handle large currents not supported by existing chip ferrite beads. Moreover, we have also prepared the **BLM18SP** and other series as a lineup of small products.

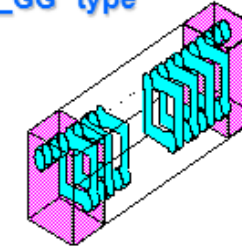


Classification of Ferrite beads

Impedance – Frequency Characteristics

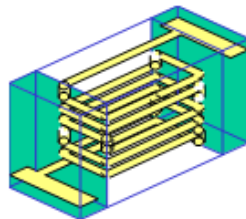


BLM_GG type



Improve of
ferrite material

BLM_AG type

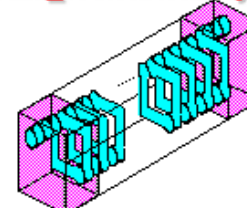


Vertical winding

Improve for
High Frequency

Frequency(MHz)

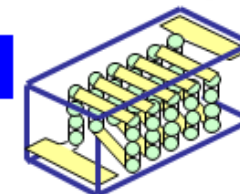
BLM_HG/HD/HB type



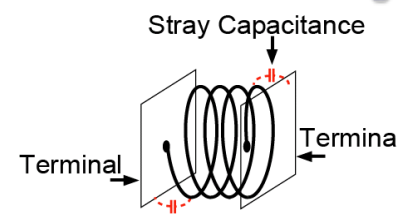
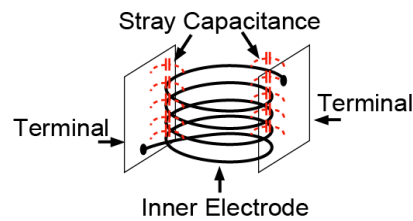
Horizontal winding

Lower Rdc

BLM_EG type



Horizontal winding
Vertical pile



Automotive grade Ferrite Bead Line-up

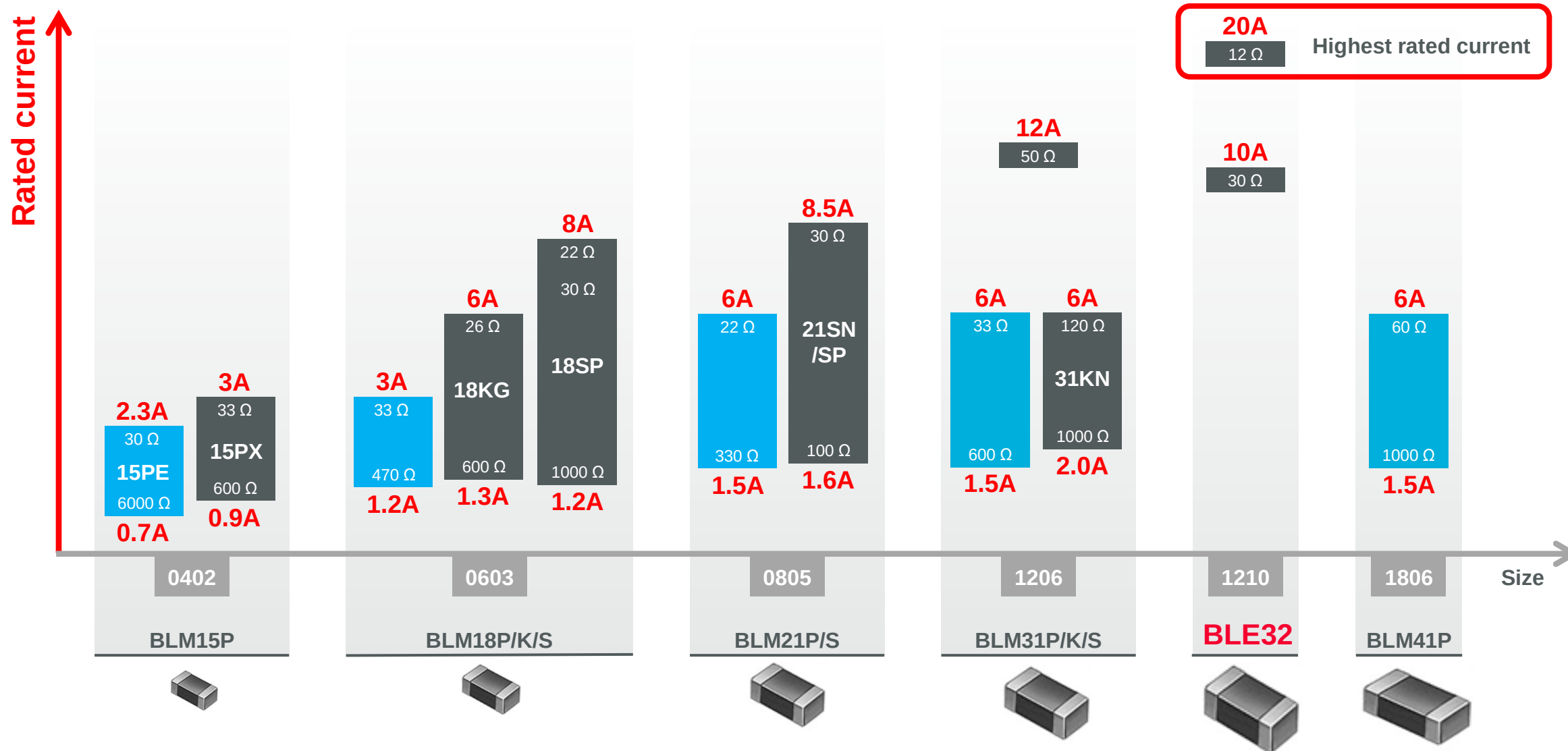
Circuit	Target Frequency	Series	0201 BLM03 	0402 BLM15 	0603 BLM18 	0805 BLM21 	1206 BLM31 	1806 BLM41 
Signal	100M to 1GHz	BLM**AG		10 to 1000Ω 0.2 to 1.0A	★ 120 to 1000Ω 0.45 to 0.8A	★ 120 to 1000Ω 0.6 to 1.0A		
		BLM**BA/BB		5 to 220Ω 0.2 to 0.5A	5 to 470Ω 0.3 to 0.8A	5 to 470Ω 0.4 to 1.0A		
		BLM**BD		470 to 1800Ω 0.2A	★ 47 to 2500Ω 0.15 to 0.5A	120 to 2700Ω 0.2 to 0.35A		
High Speed	1G to 3GHz	BLM**HG/HD	600 to 1200Ω 0.1 to 0.15A	★ 600 to 1800Ω (150C ~1000Ω) 0.2 to 0.3A	470 to 1000Ω 0.05 to 0.2A			
		BLM**HB/HE		120 to 220Ω 0.25 to 0.3A	600 to 1500Ω 0.5 to 0.8A			
	1G to 5GHz	BLM**EB	25 to 50Ω 0.4 to 0.6A					
High Speed Power	1G to 3GHz	BLM**EG		120 to 220Ω 0.7 to 1.5A	100 to 600Ω 0.5 to 2.0A			
	1G to 5GHz	BLM**DN		U.D.	150 to 600Ω 0.7 to 1.4A			
Power	100M to 1GHz	BLM**PG/PE		30 to 600Ω 0.7 to 2.3A	30 to 470Ω 1.0 to 3.0A	★ 22 to 330Ω 1.5 to 6.0A	33 to 600Ω 1.5 to 6.0A	30 to 1000Ω 1.5 to 6.0A
		BLM**PX		33 to 600Ω 0.9 to 3.0A				
		BLM**KG/KN			★ 26 to 1000Ω 1.0 to 6.0A		★ 120 to 1000Ω 2.0 to 6.0A	
		BLM**SN/SP			★ 22Ω, 8.0A	★ 30 to 1000Ω 1.6 to 8.5A	50Ω 12A	

*U.D. : under development product

*★ : 150C product available

*★ : 175C product available

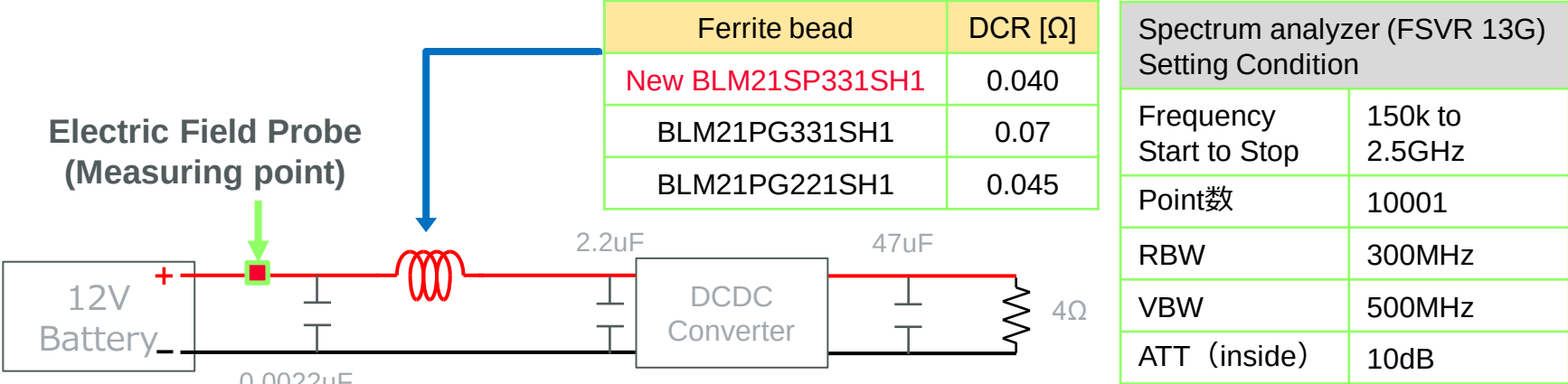
Super-high rated current



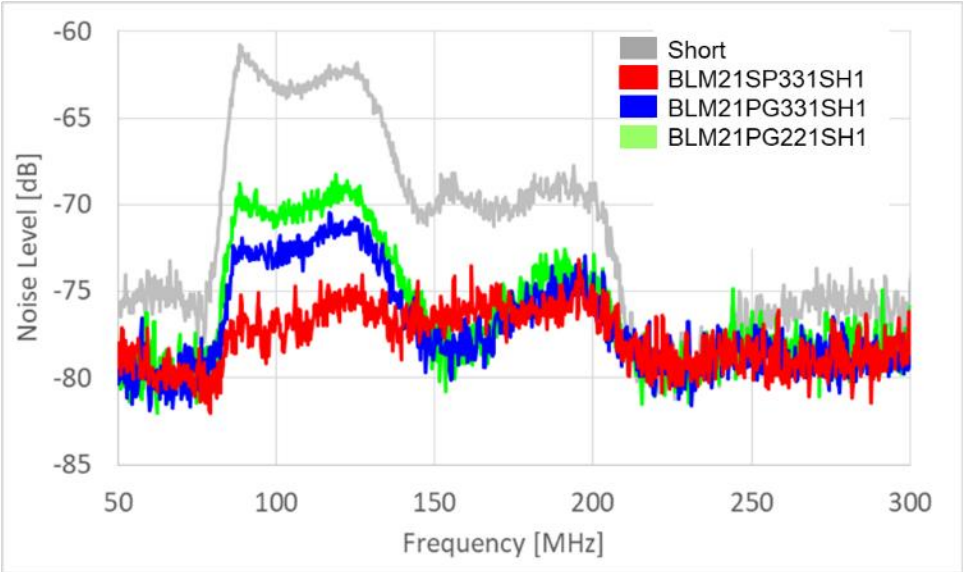
EMC performance with BLM21SP



Measurement Condition



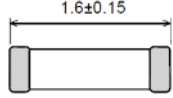
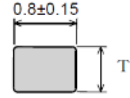
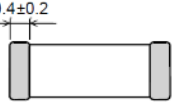
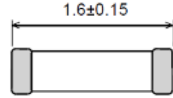
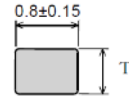
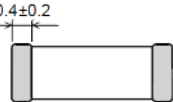
Test result



- ✓ Better noise reduction effect at FM band due to improved saturation characteristic.
- ✓ Higher rated current & lower DCR can be also realized.

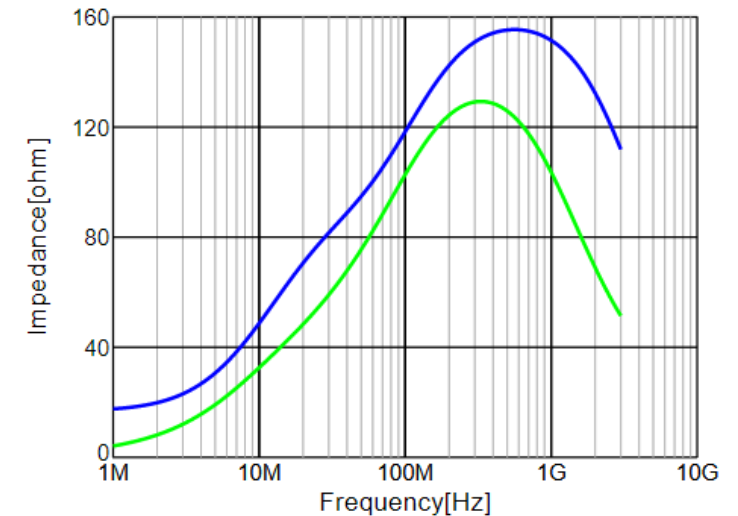
BLM18SP Series (0804 inch size) / Better than 18KG



MPN	BLM18KG101TH1	BLM18SP101SH1
Size L*W*T (mm)	1.6 x 0.8 x 0.6	1.6 x 0.8 x 0.8
	   T=0.6±0.15 : Electrode (in mm)	   T=0.8±0.15 : Electrode (in mm)
Z value @100MHz (Ω)	100±25%	100±25%
Rated current*1(A)@85°C	3.0	3.7
DC resistance (Ω)	0.03 max	0.022 max
Operating Temperature	-55°C~+125°C	-55°C~+125°C

*1 : When applied rated current to the Products, temperature rise caused by self-generated heat shall be limited to 40 max

Murata Parts Number	Impedance (Ωat100MHz)	DC Resistance (Ω max.)	Thickness (mm)	Rated Current (mA)*1	
				Up to 85C	at 125C
BLM18SP300SH1	30±10ohm	0.008	0.85±0.15	6000	4000
BLM18SP101SH1	100±25%	0.022	0.85±0.15	3700	2500
BLM18SP221SH1	220±25%	0.040	0.85±0.15	2800	1900
BLM18SP601SH1	600±25%	0.140	0.85±0.15	1500	1000
BLM18SP102SH1	1000±25%	0.185	0.85±0.15	1200	800



— BLM18SP101SH1, |Z|, DCR=0.022ohm, Rated Current=3700mA

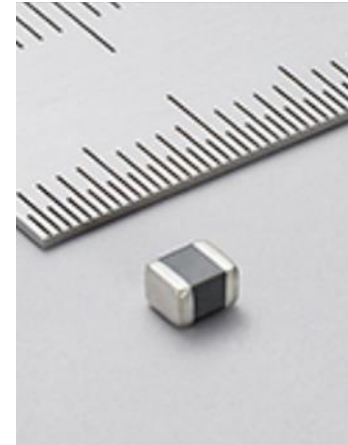
— BLM18KG101TH1, |Z|, DCR=0.03ohm, Rated Current=3000mA

Murata New Chip Ferrite bead BLE32SN

Murata has introduced the **BLE32SN** series of chip ferrite beads.

These are the world's first chip-type noise suppression components to have 20A current rating.

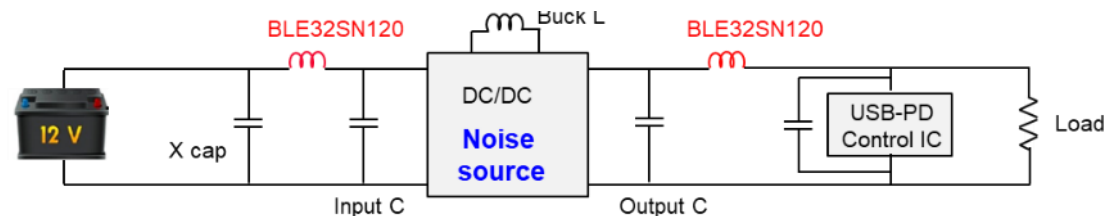
They will be used to provide noise suppression in circuits with large current flows, such as the battery charging systems and powertrains of electric vehicles (EVs), as well as in industrial equipment.



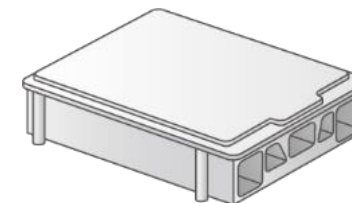
Features

- Super high rated current (20A)
- Better saturation characteristic
- High operation temperature (150C)

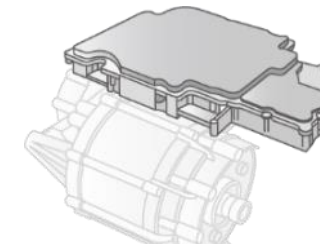
Target Circuit: Power line for xEV applications



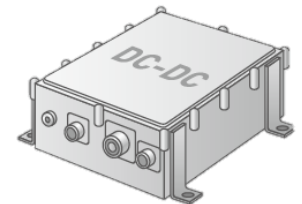
Circuit Diagram



On-board charger



Inverter



DC-DC Converter

BLE32SN Series (1210 inch size)



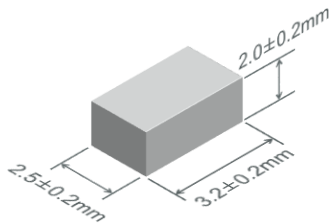
NEW PRODUCT

AEC-Q200, High Current, High impedance at low and high freq.

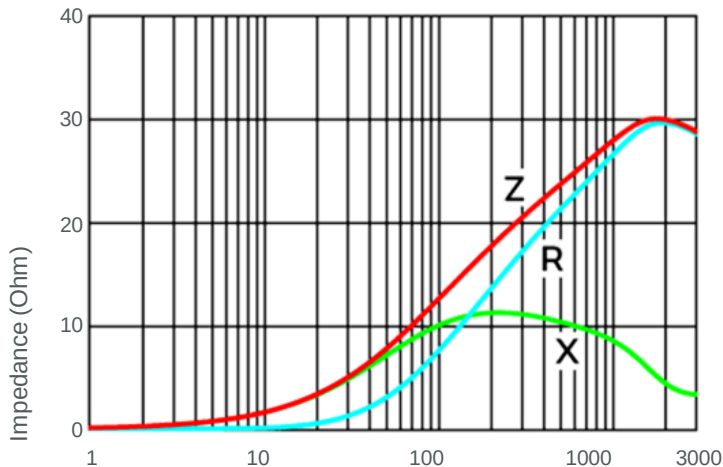
Features

- BLE32SN series is designed to function nearly as a resistor at noise frequencies, which greatly reduces the possibility of resonance and leaves signal wave forms undistorted.
- BLE32SN series can be used in high current circuits due to its low DC resistance. It can match power lines to a maximum of 20A DC.

Dimensions



Performance



Specifications

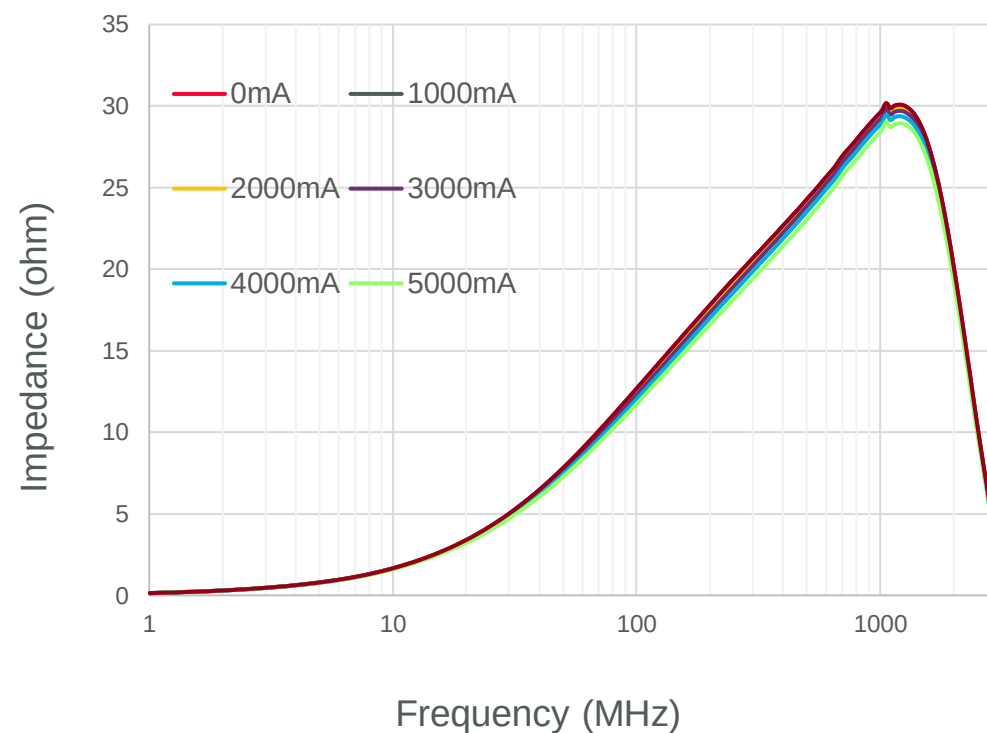
Operating temperature range: -55 ~ +125C/150C (**=SH: 125C / BH: 150C)

Murata P/N	Size LxWxH (mm)	Impedance (Ohm)	Rated Current (mA)	
		100MHz	Up to 125C	At 150C
BLE32SN120**1	3.2 x 2.5 x 2.0	12	20	0.01

Better saturation characteristics

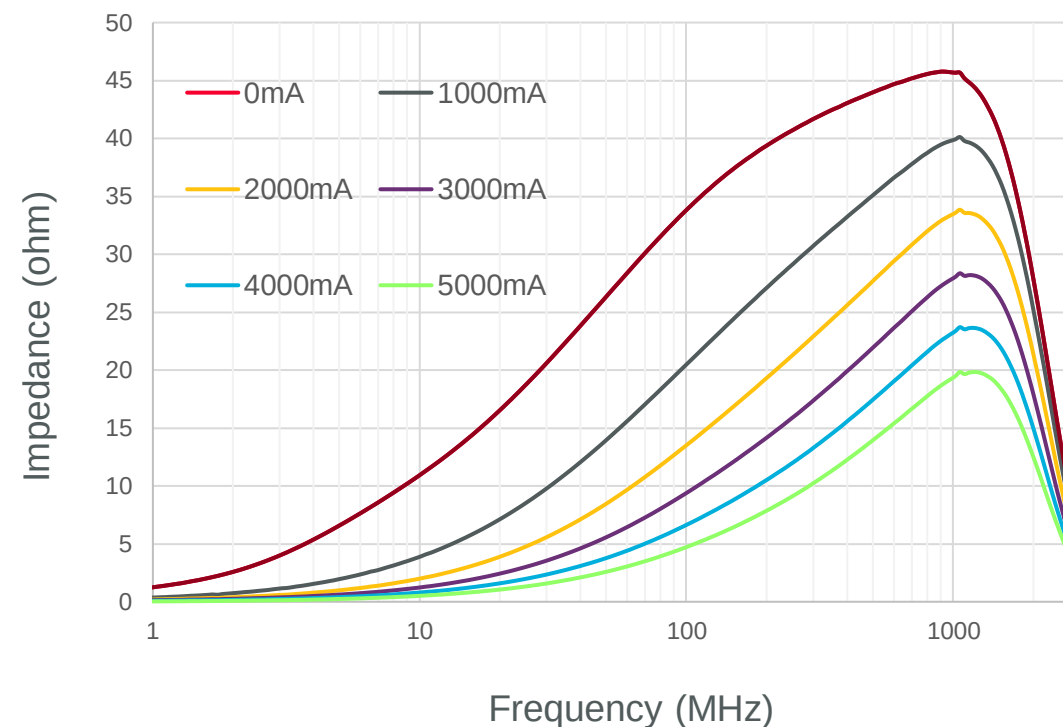
- Good saturation type

BLE32SN120



- Conventional type

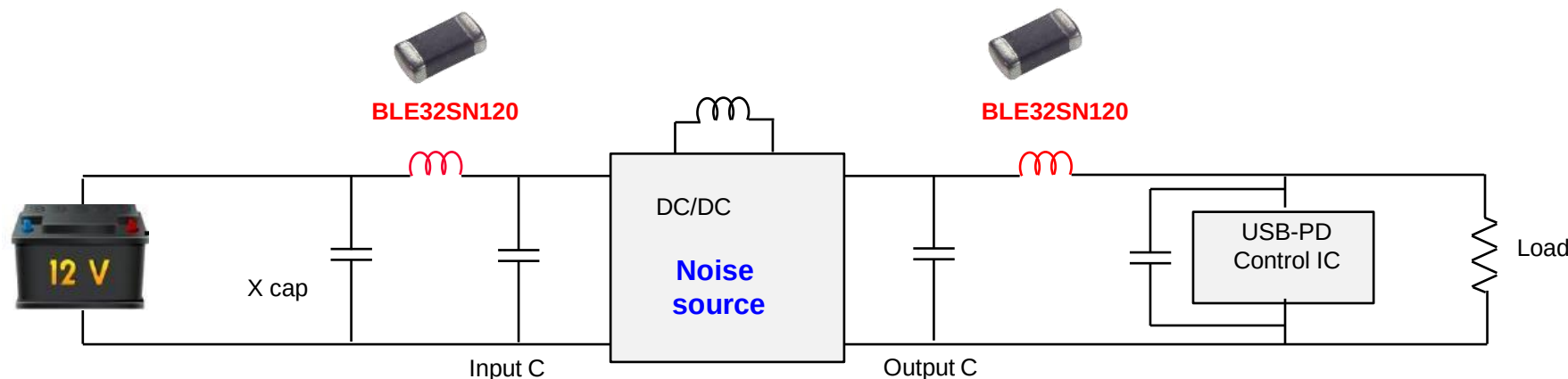
BLM31PG330



Stable Impedance value for each applied current

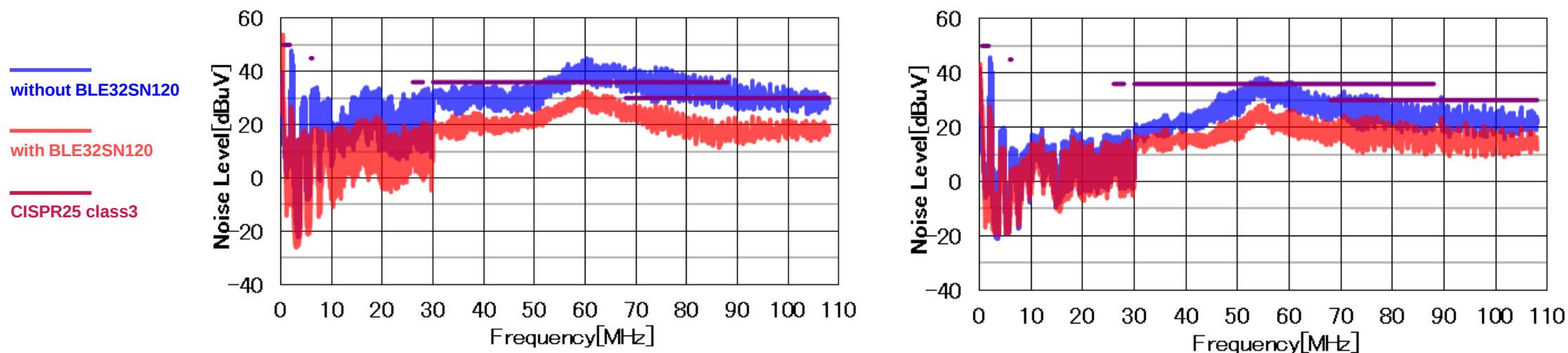
Case Study of BLE32SN120 EMC Effects

Circuit Diagram



Conducted Emission

BLE32SN120 can reduce the noise under high current condition.





***Thank you for choosing
Murata !***

